

7. SLIDE STAINER

Automated or semi-automated staining of tissue sections using validated reagent sequences.

SOP No.: LAB-HISTO-07	Revision: 01	Effective: _____
Department: Histopathology / Research	Prepared by: _____	Approved by: _____

Purpose

- To provide consistent staining of tissue sections by controlled reagent exposure, washing, differentiation and/or counterstaining according to the selected validated protocol.

PPE & Safety

- Wear lab coat, gloves and eye protection. Handle stains, solvents and reagents according to their SDS and local chemical-safety procedures. Use ventilation for volatile reagents. Treat slides as potentially infectious until appropriately processed.

Materials

- Prepared slides; staining racks; validated stains/reagents; wash solutions; reagent containers; timer/program; appropriate waste containers.

Procedure

- Verify instrument cleanliness, reagent identity, concentration, level, expiry and sequence. Confirm the correct validated staining program.
- Load labelled slides securely and select the required program. Ensure reagent containers are correctly positioned.
- Start the staining cycle. Monitor reagent levels and instrument alarms; do not interrupt the cycle without following the validated recovery procedure.
- At completion, remove slides using appropriate PPE and proceed to dehydration, clearing, mounting or the next validated step.
- Inspect representative slides for staining quality and document the run, reagent lot information and deviations where required.

Quality Control

- Assess nuclear/cytoplasmic contrast or other stain-specific endpoints using laboratory controls. Compare with an acceptable reference slide/control and repeat the run when QC fails.

Cleaning & Shutdown

- Dispose of used reagents and waste according to chemical-waste procedures. Clean racks, probes and accessible surfaces as specified by the manufacturer. Keep reagent containers closed and instrument in standby/shutdown mode as required.

Troubleshooting

- Weak staining → check reagent strength, timing and section quality. Excessive background → check differentiation/washing and reagent contamination. Uneven staining → inspect slide placement, reagent volume and agitation.

Important: This is a general SOP template. Before operational use, insert the exact manufacturer model, validated temperatures/times, reagent names, decontamination procedure, QC acceptance criteria and institutional biosafety/chemical-safety requirements.